Week report 5

The Linux File System

ims

```
pccc@cis106:~$ wget https://cis106.com/assets/practicels.tar.xz
--2026-03-09 19:24:47--  https://cis106.com/assets/practicels.tar.xz
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2229212 (2.1M) [application/x-xz]
Saving to: ‘practicels.tar.xz.1’

practicels.tar.xz.1 100%[===================>]   2.12M  9.50MB/s    in 0.2s

2026-03-09 19:24:48 (9.50 MB/s) - ‘practicels.tar.xz.1’ saved [2229212/2229212]

pccc@cis106:~$ tar Jxf practicels.tar.xz
pccc@cis106:~$ cd practicecels/
bash: cd: practicecels/: No such file or directory
pccc@cis106:~$ ls
 btw         Downloads                    Pictures              Public
 btw.zip     essentials.sh                practicels            scripts
 cis106      fonts                        practicels.tar.xz     Templates
 Desktop    'git config global user.txt'  practicels.tar.xz.1   Videos
 Documents   Music                        practiceonly          vscode.sh
pccc@cis106:~$ cd practicels/
pccc@cis106:~/practicels$ ls
book.txt  countries.xml  history-debian.pdf  homework.docx  movie.mp4  people.csv  song.mp3  tux-nobg.png  tux-penguin.gif
pccc@cis106:~/practicels$
```

```
pccc@cis106:~/practicels$ ls -lht
total 2.9M
-rw-rw-r-- 1 pccc pccc  24K Feb 27  2022 countries.xml
-rw-rw-r-- 1 pccc pccc 278K Feb 27  2022 people.csv
-rw-rw-r-- 1 pccc pccc 747K Feb 27  2022 song.mp3
-rw-rw-r-- 1 pccc pccc 1.5M Feb 27  2022 movie.mp4
-rw-rw-r-- 1 pccc pccc 160K Feb 27  2022 history-debian.pdf
-rw-rw-r-- 1 pccc pccc  21K Feb 27  2022 tux-nobg.png
-rw-rw-r-- 1 pccc pccc 1.1K Feb 27  2022 book.txt
-rw-rw-r-- 1 pccc pccc 109K Feb 22  2022 homework.docx
-rw-rw-r-- 1 pccc pccc  44K Jan 31  2020 tux-penguin.gif
pccc@cis106:~/practicels$ ls -lhgGs
total 2.9M
4.0K -rw-rw-r-- 1 1.1K Feb 27  2022 book.txt
 24K -rw-rw-r-- 1  24K Feb 27  2022 countries.xml
160K -rw-rw-r-- 1 160K Feb 27  2022 history-debian.pdf
112K -rw-rw-r-- 1 109K Feb 22  2022 homework.docx
1.5M -rw-rw-r-- 1 1.5M Feb 27  2022 movie.mp4
280K -rw-rw-r-- 1 278K Feb 27  2022 people.csv
748K -rw-rw-r-- 1 747K Feb 27  2022 song.mp3
 24K -rw-rw-r-- 1  21K Feb 27  2022 tux-nobg.png
 44K -rw-rw-r-- 1  44K Jan 31  2020 tux-penguin.gif
pccc@cis106:~/practicels$ ls -mr
tux-penguin.gif, tux-nobg.png, song.mp3, people.csv, movie.mp4, homework.docx,
history-debian.pdf, countries.xml, book.txt
pccc@cis106:~/practicels$ ls -1X
people.csv
homework.docx
tux-penguin.gif
song.mp3
movie.mp4
history-debian.pdf
tux-nobg.png
book.txt
countries.xml
pccc@cis106:~/practicels$
```

```
     --time-style=TIME_STYLE
                            time/date format with -l; see TIME_STYLE below
  -t                        sort by time, newest first; see --time
  -T, --tabsize=COLS        assume tab stops at each COLS instead of 8
  -u                        with -lt: sort by, and show, access time;
                            with -l: show access time and sort by name;
                            otherwise: sort by access time, newest first

  -U                        do not sort directory entries
  -v                        natural sort of (version) numbers within text
  -w, --width=COLS          set output width to COLS.  0 means no limit
  -x                        list entries by lines instead of by columns
  -X                        sort alphabetically by entry extension
  -Z, --context             print any security context of each file
      --zero                end each output line with NUL, not newline
  -1                        list one file per line
      --help        display this help and exit
      --version     output version information and exit

The SIZE argument is an integer and optional unit (example: 10K is 10*1024).
Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers of 1000).
Binary prefixes can be used, too: KiB=K, MiB=M, and so on.

The TIME_STYLE argument can be full-iso, long-iso, iso, locale, or +FORMAT.
FORMAT is interpreted like in date(1).  If FORMAT is FORMAT1<newline>FORMAT2,
then FORMAT1 applies to non-recent files and FORMAT2 to recent files.
TIME_STYLE prefixed with 'posix-' takes effect only outside the POSIX locale.
Also the TIME_STYLE environment variable sets the default style to use.

The WHEN argument defaults to 'always' and can also be 'auto' or 'never'.

Using color to distinguish file types is disabled both by default and
with --color=never.  With --color=auto, ls emits color codes only when
standard output is connected to a terminal.  The LS_COLORS environment
variable can change the settings.  Use the dircolors(1) command to set it.
```

```
Tilix: pccc@cis106: ~
1: pccc@cis106: ~
pccc@cis106:~$ pwd
/home/pccc
pccc@cis106:~$ cd
pccc@cis106:~$ cd /
pccc@cis106:/$ cd usr/share/themes
pccc@cis106:/usr/share/themes$ cd ../../
pccc@cis106:/usr$ ls -A ~
 .bash_aliases                  .mozilla
 .bash_history                  Music
 .bash_logout                   participationclass
 .bashrc                        Pictures
 btw                            .pki
 btw.zip                        practicels
 .cache                         practicels.tar.xz
 cis106                         practicels.tar.xz.1
 .config                        practiceonly
 Desktop                        .profile
 Documents                      Public
 .dotnet                        scripts
 Downloads                      .ssh
 essentials.sh                  .sudo_as_admin_successful
 .face                          Templates
 .face.icon                     .vboxclient-clipboard-tty2-control.pid
 fonts                          .vboxclient-clipboard-tty2-service.pid
 .gitconfig                     .vboxclient-draganddrop-tty2-control.pid
'git config global user.txt'    .vboxclient-hostversion-tty2-control.pid
 .gnupg                         .vboxclient-seamless-tty2-control.pid
 lab5files                      .vboxclient-vmsvga-session-tty2-control.pid
 lab5files.zip                  .vboxclient-vmsvga-session-tty2-service.pid
 list                           Videos
 .local                         .vscode
 marstenHouse                   vscode.sh
 marstenhouse.sh                .wget-hsts
pccc@cis106:/usr$ cd; ls -tX cis106/
cheatsheets  Final_project  labs  notes  week_reports
pccc@cis106:~$ ls -rR ~/cis106/
/home/pccc/cis106/:
week_reports  notes  labs  Final_project  cheatsheets

/home/pccc/cis106/week_reports:
wr5  wr4  wr3  wr2  wr1

/home/pccc/cis106/week_reports/wr5:
wr5.md  fspractie1.png  fspractice2b.png  fspractice2a.png

/home/pccc/cis106/week_reports/wr4:
wr4.png  wr4.pdf  wr4.md  wr4.jpeg  wr4.html  p1shell.png  p1bastet.png

/home/pccc/cis106/week_reports/wr3:
wr3.md          practice8.png  practice6.png  practice4.png  practice10error.png
practice9.png  practice7.png  practice5.png  practice3.png

/home/pccc/cis106/week_reports/wr2:
wr2.pdf  wr2.md  debiandesktop.png

/home/pccc/cis106/week_reports/wr1:
wr1.pdf  wr1.md  slack.png  github.png

/home/pccc/cis106/notes:
```

```
2: pccc@cis106: ~
  -v                         natural sort of (version) numbers within text
  -w, --width=COLS           set output width to COLS.  0 means no limit
  -x                         list entries by lines instead of by columns
  -X                         sort alphabetically by entry extension
  -Z, --context              print any security context of each file
      --zero                 end each output line with NUL, not newline
  -1                         list one file per line
      --help        display this help and exit
      --version     output version information and exit

The SIZE argument is an integer and optional unit (example: 10K is 10*1024).
Units are K,M,G,T,P,E,Z,Y,R,Q (powers of 1024) or KB,MB,... (powers of 1000).
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then FORMAT1 applies to non-recent files and FORMAT2 to recent files.
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with --color=never.  With --color=auto, ls emits color codes only when
standard output is connected to a terminal.  The LS_COLORS environment
variable can change the settings.  Use the dircolors(1) command to set it.

Exit status:
 0  if OK,
 1  if minor problems (e.g., cannot access subdirectory),
 2  if serious trouble (e.g., cannot access command-line argument).

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/ls>
or available locally via: info '(coreutils) ls invocation'
pccc@cis106:~$
```
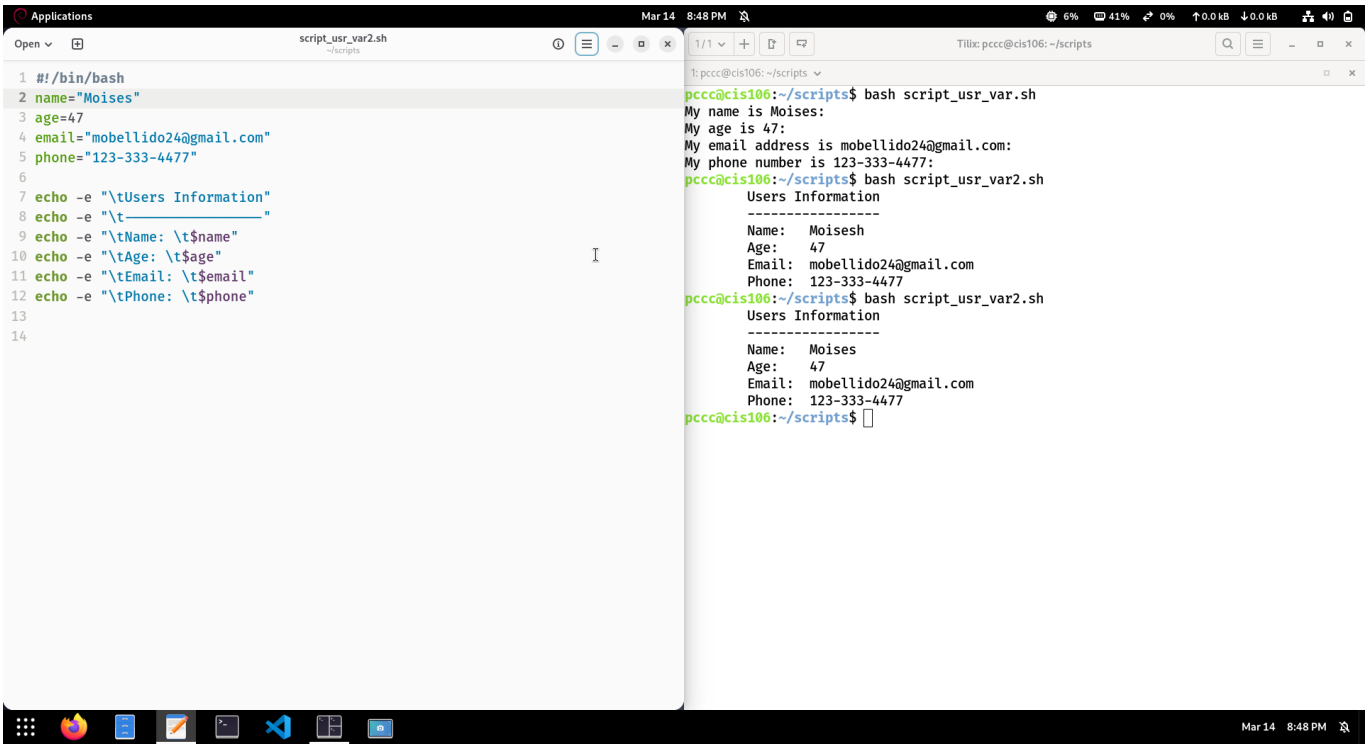

```
pccc@cis106: ~
drwxr-xr-x 4 pccc pccc 4.0K Mar 12 18:24  Downloads
-rwxrwxr-x 1 pccc pccc 7.6K Feb 19 10:19  essentials.sh
drwxrwxr-x 2 pccc pccc 4.0K Feb 21 11:01  fonts
-rw-rw-r-- 1 pccc pccc  233 Feb 20 23:17 'git config global user.txt'
drwxrwxr-x 2 pccc pccc 4.0K Feb 22  2022  lab5files
-rw-rw-r-- 1 pccc pccc 3.5M Mar  6 13:38  lab5files.zip
-rw-r--r-- 1 pccc pccc 9.4K Mar 11 20:04  list
drwxrwxr-x 3 pccc pccc 4.0K Mar 11 21:29  marstenHouse
-rw-rw-r-- 1 pccc pccc 3.4K Mar  6 13:38  marstenhouse.sh
drwxr-xr-x 2 pccc pccc 4.0K Feb 20 22:14  Music
drwxrwxr-x 2 pccc pccc 4.0K Feb 26 21:12  participationclass
drwxr-xr-x 4 pccc pccc 4.0K Mar 12 18:27  Pictures
drwxrwxr-x 2 pccc pccc 4.0K Feb 27  2022  practicels
-rw-rw-r-- 1 pccc pccc 2.2M Mar  6 13:38  practicels.tar.xz
-rw-rw-r-- 1 pccc pccc 2.2M Mar  6 13:38  practicels.tar.xz.1
drwxrwxr-x 2 pccc pccc 4.0K Mar 12 18:17  practiceonly
drwxr-xr-x 2 pccc pccc 4.0K Feb 20 22:14  Public
drwxrwxr-x 2 pccc pccc 4.0K Mar 12 18:09  scripts
drwxr-xr-x 2 pccc pccc 4.0K Feb 20 22:14  Templates
drwxr-xr-x 3 pccc pccc 4.0K Mar 12 18:27  Videos
-rwxrwxr-x 1 pccc pccc 2.5K Feb 19 10:19  vscode.sh
pccc@cis106:~$ ls -lGg --full-time ~
total 19916
drwxrwxr-x 3     4096 2026-02-22 13:01:00.299511163 -0500  btw
-rw-rw-r-- 1 12253838 2026-02-22 11:35:56.169353329 -0500  btw.zip
drwxrwxr-x 8     4096 2026-03-08 22:35:28.381582985 -0400  cis106
drwxr-xr-x 2     4096 2026-02-20 22:14:49.416000846 -0500  Desktop
drwxr-xr-x 5     4096 2026-03-12 18:27:29.158991475 -0400  Documents
drwxr-xr-x 4     4096 2026-03-12 18:24:27.466989555 -0400  Downloads
-rwxrwxr-x 1     7747 2026-02-19 10:19:26.000000000 -0500  essentials.sh
drwxrwxr-x 2     4096 2026-02-21 11:01:53.653125392 -0500  fonts
-rw-rw-r-- 1      233 2026-02-20 23:17:54.672122199 -0500 'git config global user.txt'
drwxrwxr-x 2     4096 2022-02-22 14:18:06.000000000 -0500  lab5files
-rw-rw-r-- 1  3570561 2026-03-06 13:38:22.000000000 -0500  lab5files.zip
-rw-r--r-- 1     9567 2026-03-11 20:04:46.627077398 -0400  list
drwxrwxr-x 3     4096 2026-03-11 21:29:32.268990338 -0400  marstenHouse
-rw-rw-r-- 1     3407 2026-03-06 13:38:23.000000000 -0500  marstenhouse.sh
drwxr-xr-x 2     4096 2026-02-20 22:14:49.416000846 -0500  Music
drwxrwxr-x 2     4096 2026-02-26 21:12:19.262496303 -0500  participationclass
drwxr-xr-x 4     4096 2026-03-12 18:27:29.158991475 -0400  Pictures
drwxrwxr-x 2     4096 2022-02-27 13:46:18.000000000 -0500  practicels
-rw-rw-r-- 1  2229212 2026-03-06 13:38:23.000000000 -0500  practicels.tar.xz
-rw-rw-r-- 1  2229212 2026-03-06 13:38:23.000000000 -0500  practicels.tar.xz.1
drwxrwxr-x 2     4096 2026-03-12 18:17:41.214985263 -0400  practiceonly
drwxr-xr-x 2     4096 2026-02-20 22:14:49.416000846 -0500  Public
drwxrwxr-x 2     4096 2026-03-12 18:09:14.919342678 -0400  scripts
drwxr-xr-x 2     4096 2026-02-20 22:14:49.416000846 -0500  Templates
drwxr-xr-x 3     4096 2026-03-12 18:27:29.158991475 -0400  Videos
-rwxrwxr-x 1     2501 2026-02-19 10:19:26.000000000 -0500  vscode.sh
pccc@cis106:~$
```

Shell Script Chapter 2

```
pccc@cis106:~/scripts$ bash script_env_var.sh
The current user: pccc
The current shell: /bin/bash
The current's user Home: /home/pccc
The Path Var:
 /usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/snap/bin:/usr/bin
pccc@cis106:~/scripts$
```

```
pccc@cis106:~/scripts$ bash script_usr_var.sh
My name is Moises:
My age is 47:
My email address is mobellido24@gmail.com:
My phone number is 123-333-4477:
pccc@cis106:~/scripts$
```


Applications
Mar 14 8:48 PM
Open
script_usr_var2.sh
~/scripts
#!/bin/bash
name="Moises"
age=47
email="mobellido24@gmail.com"
phone="123-333-4477"
echo -e "\tUsers Information"
echo -e "\t————————————————"
echo -e "\tName: \t$name"
echo -e "\tAge: \t$age"
echo -e "\tEmail: \t$email"
echo -e "\tPhone: \t$phone"
Tilix: pccc@cis106: ~/scripts
1: pccc@cis106: ~/scripts
pccc@cis106:~/scripts$ bash script_usr_var.sh
My name is Moises:
My age is 47:
My email address is mobellido24@gmail.com:
My phone number is 123-333-4477:
pccc@cis106:~/scripts$ bash script_usr_var2.sh
Users Information

Name: Moisesh
Age: 47
Email: mobellido24@gmail.com
Phone: 123-333-4477
pccc@cis106:~/scripts$ bash script_usr_var2.sh
Users Information

Name: Moises
Age: 47
Email: mobellido24@gmail.com
Phone: 123-333-4477
pccc@cis106:~/scripts$
Mar 14 8:48 PM